Predictive Modeling of Information Flow Across Brain Regions on Multi-Scale Level

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Abstract

Understanding how information flows between brain regions is essential for uncovering the neural basis of behavior. Using the Neuropixels dataset of mice performing a visual discrimination task, we investigated whether activity in one brain region can predict firing patterns in downstream targets. At the population level, we modeled normalized total spike counts across sessions using LASSO-regularized GLMs with EM-imputed regional activity. We found that midbrain activity was most predictable, followed by hippocampus and basal ganglia, with thalamic and visual areas serving as key contributors. At the neuron level, we trained log-linear neural models using frequency-domain features of single-neuron spike trains. Despite variability across neurons and sessions, the model still captured some relevant patterns. Together, these approaches offer complementary insights into functional connectivity across spatial and temporal scales, and highlight how sensory and modulatory regions contribute to downstream dynamics during decision-making tasks.

1 Introduction

Understanding information flow in the brain is a central goal in neuroscience, as it reveals how neural circuits process, transmit, and transform sensory inputs and motor commands to generate behavior. This knowledge is crucial for interpreting the computations performed by the brain, diagnosing and treating neurological disorders, and designing interventions that can modulate neural processing [1]. Modern approaches, including neural information flow models and deep learning frameworks, now enable researchers to estimate and visualize these complex, distributed processes directly from large-scale neural data, offering biologically interpretable insights into how information is encoded, transformed, and communicated across interconnected brain regions.

In this study, we analyze a Neuropixels dataset collected during a visual decision-making task in mice to investigate how information flows across visual, thalamic, midbrain, hippocampal, and basal ganglia regions. Using both population ROI-level and neuron-level modeling approaches, we quantify how well upstream activity can predict downstream responses, aiming to understand functional connectivity patterns and task-relevant modulation across brain structures.

2 Background and Related Work

Most of the literature use non-deep learning methods to study the relay of information. For instance, Bilek et al. [2] used hyperscanning EEG and cross-correlation analyses to study neural coupling between people engaged in social interaction and found time-lagged coupling between regions such as the right temporoparietal junction to medial prefrontal cortex. At the population level, Generalized Linear Models (GLMs) have been widely employed to study inter-regional dependencies. Kobayashi et al. [3] modeled the firing probability of a postsynaptic neuron as a function of presynaptic spiking, fitting GLMs to cross-correlations between spike trains in simulated data. Similarly, Noel et al. [4] used a coupled point-process generalized additive model (P-GAM) to study neural dynamics during

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a virtual navigation task, modeling spikes as a function of task variables and local field potentials across cortical areas.

At the systems level, Steinmetz el al. [5] used joint Peri-Event Cross-Correlation (jPECC) to analyze the temporal relationships between activity patterns in different brain areas. They found that 98% choice-selective neurons preferred contralateral choices in the midbrain. On the contrary, choice-selective neurons in the forebrain could prefer either choice, with 40% preferring ipsilateral choices. It seems that the midbrain is more task-specific and the forebrain is more integrative, which motivates our investigation of using a combination of inputs to predict the neural activity in downstream brain regions.

While deep learning methods are emerging in this space, their application remains limited. For example, Seeliger et al. [6] introduced Neural Information Flow models using coupled convolutional networks to map hierarchical interactions across regions, though these approaches have not yet been applied at the single-neuron level. Other efforts have focused on anatomical and temporal aspects of neural computation. Recent work such as that by Li et al. [7] integrates anatomical information (e.g., axonal structures) with neural networks to infer brain connectivity. In the work by Wang et al. [8], a Recurrent Neural Network (RNN) is employed to predict future EEG signals, which were then used to compute Granger Causality. While RNNs are commonly used for modeling time-series neural activity, there has been limited exploration of combining GLMs with Convolutional Neural Networks (CNNs) in this context. However, RNNs typically require large datasets for effective training, making them less suitable for the present study.

Our study combines the strengths of interpretable statistical models and compact spectral representations to examine brain-wide information flow, with additional explorations into the current gaps in using deep learning methods to study information flow. We adopt a two-part strategy: population-level GLMs to assess region-to-region influences, and neuron-level spectral models to capture finer temporal dynamics.

3 Proposal

3.1 Dataset

We used the Neuropixels dataset from Steinmetz et al. [5], which includes 30,000 neurons across 42 brain areas recorded during mice performing a visual discrimination task. This task provides movement versus no-movement responses, allowing for straightforward dataset segmentation and targeted deep learning model design. The dataset provides trial-aligned neuron spikes, anatomical labels, and behavioral signals (e.g., stimulus contrast, response direction, feedback). Considering that it also includes different stimuli and types (e.g., category of the task, i.e., stimulus occurring on the left or right side, the type of visual stimulus, and/or the involvement of motor movements), we were also able to explore such covariates through directional (left/righ stimuli) and categorical embeddings. Below are population-level temporal spike plots from this dataset provided by the authors, which we use to select the ROIs for our study (visual cortex, thalamus, hippocampus, and other cortical regions).

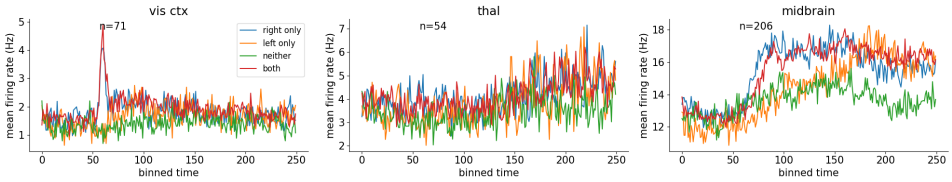


Figure 1: Population average of different ROI based on different task condition

3.2 Implementation

After initial exploratory data analysis, we found that there is limited session with recording from all of our desired ROIs. This lead us to split our methodology into two parts: one focusing on population-level modeling using region-averaged activity and GLMs to quantify how well different input sources can predict average activity in downstream targets; the other focusing on using existing

individual neurons from any sub-region in the source region to predict the time series of individual neurons from the target region.

To address incomplete coverage across brain regions, we also explored missing data imputation methods to approximate whole-brain activity, enabling a broader examination of ROI contributions during task performance. These exploratory efforts complemented our main analysis, which focused on targeted models using activity from a single input source (e.g., visual cortex, thalamus, or midbrain) to predict responses in three downstream regions: hippocampus, midbrain, and basal ganglia, either on a population-level or neuron-level. **We hypothesize that the prediction performance provides indirect evidence of information flow. From the conclusion drawn by Steinmetz et al. [5], we expect a better performance in predicting midbrain activity, seeing that there are more choice-specific neurons and may be more task-specific.**

3.2.1 Population Level Analysis

For our population-level analysis, we computed normalized total spike counts for each ROI in each session. Let x_n denote the total spike count of neuron $n \in \mathcal{N}_{r,s}$ over the entire session duration. The normalized spike count for ROI r in session s is calculated as:

$$X_{r,s} = \frac{1}{|\mathcal{N}_{r,s}|} \sum_{n \in \mathcal{N}_{r,s}} x_n.$$

These values $X_{r,s}$ were then used as covariates or targets in downstream imputation and regression models.

Due to limited simultaneous recordings across all ROIs, we employed a Gaussian Expectation-Maximization (EM) matrix completion method to impute missing regional activity. The input matrix $X \in \mathbb{R}^{T \times D}$ contained normalized total spike counts, where each row represents a session and each column an ROI. Missing values were marked as NaNs. The EM algorithm assumes a multivariate Gaussian distribution and iteratively: (1) estimates the mean vector $\boldsymbol{\mu}$ and covariance matrix Σ from the current matrix; and (2) imputes missing values in each row using conditional expectations based on observed entries. We z-scored the matrix for numerical stability, initialized missing values with column-wise means, and iterated until convergence. The final imputed matrix was transformed back to the original scale. To avoid information leakage, any ROIs used as targets in a given analysis were excluded from both the imputation process and the final predictor matrix.

We modeled normalized spike counts in each target ROI using GLM with a log link. To satisfy GLM assumptions, both preprocessed predictors and targets were shifted to be non-negative integers. While Poisson regression was the initial choice, we used Negative Binomial (NB) models in cases of overdispersion. We split the data into 80% training and 20% testing sets to evaluate out-of-sample performance. To prevent overfitting and interpret fine-grained contributions, we applied LASSO with 5-fold cross-validation to select a subset of informative sub-ROIs from different input matrices. GLM was then fit using only these selected predictors, along with task-related covariates. LASSO was implemented with `sklearn`, and GLM was implemented with `statsmodels.api`.

In addition to neural predictors, we included trial-level task variables to account for sensory and decision context. These covariates included contrast levels on the left and right screens, and a binary no-go indicator reflecting whether the mouse failed to respond during the trial, potentially indicating disengagement.

To systematically study information flow between anatomically and functionally distinct brain regions, we conducted four sets of GLM-based analyses. The first two constitute the core of our population-level modeling: (1) a whole-cortex-to-target model characterizing distributed contributions across the brain, and (2) a visual-cortex-to-target model isolating feedforward sensory-driven predictions. These were followed by two supplemental analyses designed to explore potential relay and modulatory mechanisms: (3) a combined visual and thalamic input model assessing how sensory relay nuclei shape downstream activity, and (4) midbrain-to-hippocampus/basal ganglia models probing whether midbrain activity, linked to behavioral salience and gating, helps explain target dynamics.

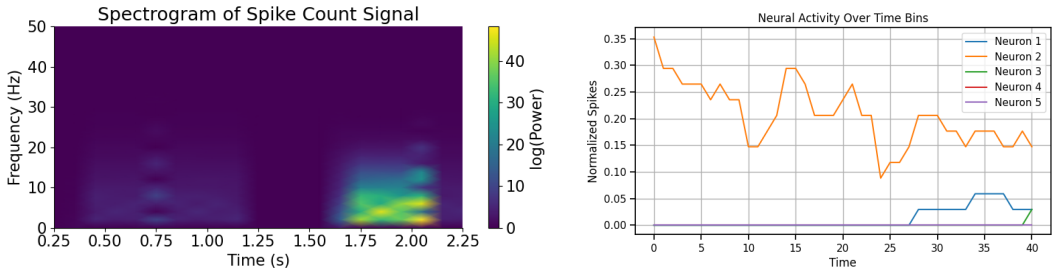
All analyses followed a unified pipeline: Gaussian EM imputation was first performed on all available ROIs excluding the target region to obtain a complete covariate matrix; LASSO-based feature selection was then used to identify informative source ROIs; finally, a GLM with either Poisson or NB likelihood was fit using the selected source features along with task-related covariates. The

predictive performance of these models offers a comparative readout of functional influence, while the selected ROIs and their coefficients provide interpretable clues about candidate pathways of neural communication. Together, these analyses construct a high-level map of region-to-region influence across the brain and provide a foundation for the next stage of analysis, which shifts to a finer temporal and spatial scale that models information flow between individual neurons during task performance.

3.2.2 Neuron Level Analysis

This approach focuses on using multiple single-neuron time-series data from the visual region to predict neuronal spiking activity in the target region. Given the large and variable number of neurons across mice, we employed heuristic strategies for source neuron selection. These included random selection, selecting neurons with the highest spiking standard deviation (under the assumption that greater variability may reflect increased stimulus responsivity and task engagement), and using the absolute difference between maximum and minimum spike counts as a proxy for dynamic range.

A total of 39 sessions were first filtered to include only sessions that had data available in at least one sub-region from both the source and target regions per session. One session was held out for testing, while the remaining sessions were split into 90% training and 10% validation set. For input features, we extracted Short-Time Fourier Transform (STFT) representations using `SciPy` library with a 50ms time bin and 40ms overlap, followed by temporal smoothing with a Gaussian kernel (σ = 10ms). The choice of time bin affects frequency resolution, e.g., a 30ms bin captures fewer frequency components than a 50ms bin. To enhance contrast, a logarithmic transform was applied to the spectral power, and all input features were normalized to [0, 1] range to ensure Poisson loss validity. The training pipeline was adapted from Lab 4, with a reduced learning rate. The final model had an input shape of (batch, timestep, num_neurons_per_ROI × num_ROIs, num_frequencies) and an output of shape (batch, timestep, num_target_neuron), where batches were consisted of vertically stacked trials in the training or validation sessions. Poisson log-likelihood was used as the loss function.



(a) Example of spectral input

(b) Example of target

Figure 2: Example of spectral input and targets

We also experimented with adding conditional embeddings into the LNP model, including trial-specific features such as left/right contrast difference and laterality. However, due to a strong class imbalance (contralateral trials outnumbering ipsilateral trials by a factor of four), these embeddings did not yield significant performance improvements. We further evaluated an LNP variant with 2D convolutional layers, but similarly observed no substantial gains. This is likely attributable to the limited size of the dataset, which may not support the effective training of models with a larger number of parameters.

3.2.3 Evaluation Metrics

For the population-level analysis, we assessed model performance using standard regression metrics: mean squared error (MSE) and coefficient of determination (R^2) on held-out test data. For GLMs, we also report pseudo-R^2, which compares the log-likelihood of the fitted model to a null model with intercept only. For neuron-level time series predictions, in addition to using normalized mean-square error (NMSE), we used dynamic time warping (DTW) to capture the similarity of the time-series prediction. DTW has been widely used to compare and classify neural spike waveforms [9]. In time series prediction (including neural data), DTW can be used to evaluate prediction performance by

quantifying the similarity between predicted and observed firing patterns, especially when temporal misalignments occur [10].

4 Results and Discussion

4.1 Population-Level Spikes Modeling

Table 1: Model performance across target regions using Poisson and NB GLMs.

Target	GLM Type	Input	MSE	Pseudo R^2 (CS)	R^2	Selected ROIs
Midbrain	Poisson	Task Only	1391.24	0.3343	0.0136	NA
Midbrain	Poisson	Visual+Task	1311.84	0.9017	0.0699	6
Midbrain	Poisson	Full+Task	1167.24	0.9935	0.1724	13
Midbrain	NB	Visual+Task	1318.75	0.0504	0.0650	6
Midbrain	NB	Full+Task	1241.28	0.1035	0.1199	13
Hippocampus	Poisson	Task Only	139.58	0.0432	0.0042	NA
Hippocampus	Poisson	Visual+Task	124.10	0.5977	0.1146	6
Hippocampus	Poisson	Full+Task	116.92	0.7325	0.1658	9
Hippocampus	NB	Visual+Task	126.64	0.0545	0.0965	6
Hippocampus	NB	Full+Task	119.36	0.0772	0.1485	9
Basal Ganglia	Poisson	Task Only	313.29	0.1633	-0.0008	NA
Basal Ganglia	Poisson	Visual+Task	303.13	0.7622	0.0316	5
Basal Ganglia	Poisson	Full+Task	289.23	0.8441	0.0760	12
Basal Ganglia	NB	Visual+Task	304.36	0.0650	0.0277	5
Basal Ganglia	NB	Full+Task	306.02	0.0851	0.0224	12

Table 2: LASSO-selected ROIs with coefficients from full-brain NB GLMs. Only statistically significant coefficients ($p < 0.05$) are included.

Target	ROI	Coef	p-value	Interpretation
Midbrain	LH	6.7598	<0.001*	Limbic arousal and relay (thalamus)
	LP	-5.0392	0.012*	Visual thalamic relay suppression
	MD	2.3577	<0.001*	Prefrontal-thalamic loop activation
	PO	35.7163	<0.001*	Somatosensory thalamic drive
	POL	14.0751	<0.001*	Multisensory thalamic boost
	AUD	-14.1716	<0.001*	Suppression of auditory input
	ILA	40.8931	<0.001*	Top-down prefrontal modulation
	MOs	-10.5751	<0.001*	Motor cortex disengagement
	PIR	42.7226	<0.001*	Olfactory and limbic excitation
	TT	-11.7841	<0.001*	Olfactory suppression
	ACB	-63.9779	<0.001*	Striatal inhibition (basal ganglia)
	SI	1.6450	0.001*	Somatosensory facilitation
Hippocampus	VISl	-0.1259	0.009*	Visual input dampening (visual cortex)
	VISp	0.2096	<0.001*	Primary visual cortex drive
	LGd	0.1974	<0.001*	Thalamic visual relay activation
	MOs	-0.0897	<0.001*	Motor cortex suppression
	TT	-0.0931	0.001*	Olfactory-limbic deactivation
Basal Ganglia	CA	-0.5520	0.001*	Hippocampal suppression
	COA	0.1298	0.004*	Olfactory-cortical drive
	DP	-0.3508	0.003*	Prefrontal suppression
	ZI	-0.1051	0.016*	Subthalamic inhibition

Table 1 summarizes model performance across different configurations. Across all three target regions, models that combined LASSO-selected full-brain inputs with task covariates performed best

under Poisson GLMs. The midbrain was the most predictable ($R^2 = 0.17, n = 4100$), followed by the hippocampus ($R^2 = 0.16, n = 6089$), and then the basal ganglia ($R^2 = 0.08, n = 4041$). This suggests that the midbrain receives highly structured input during task performance, which aligns with prior findings that it is a key integrative hub for choice- and salience-related signals [5].

However, once overdispersion was accounted for using NB GLMs, model performance dropped, especially for the midbrain and basal ganglia. Overdispersion, in this context, refers to the observation that the variance in total spike counts across trials was significantly greater than the mean, violating the core Poisson assumption of equal mean and variance. This suggests that while input patterns to these regions are structured, there exists meaningful trial-by-trial variability, possibly due to internal state changes, behavioral noise, or latent modulatory factors, that Poisson models tend to overfit. The midbrain's high variance is consistent with its modulatory role and behavioral flexibility. For the basal ganglia, the drop in fit supports the idea that its output is more thresholded or stochastic, likely reflecting downstream action selection. The hippocampus seemed more stable across GLM types, which fits its role as a conditional memory buffer being engaged during task-relevant decisions but less driven by noisy or modulatory input.

As for task covariates, the results are attached in Table 4 in the appendix. Overall, contrast level on the left and right had small but sometimes significant effects. Notably, no-go trials were associated with consistent negative coefficients across all targets and models, suggesting widespread suppression of target region activity during missed trials, which potentially reflects reduced engagement or decision-related signaling.

Table 2 on ROI coefficients in the GLM models offers interpretable insights into information pathways. For the midbrain, upstream thalamic and olfactory structures (e.g., POL, PIR, ILA) were strong positive contributors, consistent with top-down modulation and sensory convergence. Hippocampal predictors included visual and thalamic relay areas (e.g., VISp, LGd), highlighting structured sensory input. In basal ganglia models, both cortical (e.g., COA) and subthalamic (e.g., ZI) sources played modest but significant roles. In the models with visual cortex source input only, LASSO nearly selects all visual regions as informative predictors, which motivates the following targeted follow-up.

Going beyond full brain models that guides high level understanding of information flow, we further explored whether adding thalamic relay regions to visual area sources would improve predictions of downstream targets. The results are detailed in Table 5 in the appendix. This addition consistently boosted model performance when using NB GLMs, especially for midbrain and hippocampus, suggesting that thalamic relays do more than transmit sensory input. Specific subregions like mediodorsal (MD; goal-directed behavior) and lateral geniculate (LGd; visual integration) contributed uniquely to different targets.

We also examined whether midbrain activity itself could account for hippocampal and basal ganglia dynamics, as it is often associated with behavioral salience and motivational gating. The results are again detailed in Table 6 in the appendix. Models using only midbrain subregions predicted hippocampal activity comparably well ($R^2 \approx 0.15, n = 6089$), with significant contributions in regions like SCm and RN involved in spatial attention and motor coordination. However, performance dropped sharply for basal ganglia ($R^2 \approx 0.017, n = 4041$). These results hint at a functional hierarchy: thalamic and cortical inputs offer structured, feedforward drive, while the midbrain acts more as a variable, modulatory influence.

Despite offering useful insights, our population-level modeling comes with several limitations. While EM imputation helped mitigate sparse ROI coverage, session-to-session variability and limited overlap of sub-regions across mice restrict our ability to make generalizable claims about fine-grained spatial contributions. The GLMs assume independence across observations, which may not strictly hold given repeated trials within animals. Additionally, by aggregating spike counts across time, we lose access to temporal structure, making it difficult to examine time-lagged influences or dynamic signal propagation, both of which are essential to understanding neural communication. Although future extensions could involve modeling task-ROI interactions or clustering regions by coefficient patterns to uncover functional modules given task conditions, such analyses are constrained by data availability. To address these gaps and move toward a more temporally resolved view of information flow, we next shift to a neuron-level modeling framework that leverages per-trial spike dynamics in the frequency domain.

4.2 Neuronal Time Series Prediction

During the neuron-level analysis, initial experiments showed that training a CNN did not yield significantly better performance than the simpler LNP model. As a result, we proceeded with LNP for our experiments.

Across different learning rates, with an example shown in Figure 3, we observed that training losses plateaued quickly, suggesting a potential bottleneck. Interestingly, training loss was often higher than validation loss, and the reduction in loss across epochs was minimal. This is likely due to the high variability introduced by selecting source and target neurons from different sub-populations across sessions. Additionally, the selected target neuron varied significantly across mice, further contributing to inter-animal variability.

We explored several alternatives to improve model performance, including different neuron selection strategies, changes to the maximum delay (max_delay), and variations in spectrogram window size. However, none of these adjustments led to meaningful performance gains. The current session-based train/validation/test split may also be suboptimal, as intra-mouse variability might exceed inter-session variability. In practice, neural network models are often fine-tuned for individual subjects with personalized inputs, suggesting that including all sessions during training may better reflect real-world applications.

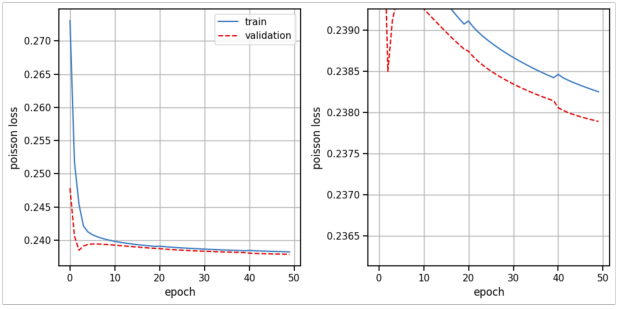


Figure 3: An example learning curve for LNP model.

Qualitatively, as shown in Figure 4, some neurons were predicted with reasonable accuracy. However, neurons without firing activity were often assigned a baseline signal, likely reflecting noise learned by the model. Only one or two neurons from the target ROI in each test session showed a clearly accurate temporal pattern, which are some details that may not be fully captured by the aggregate metrics.

Table 3: Comparison of neuron selection methods based on normalized MSE and averaged DTW.

Neuron Selection Method	Source	Target	Num Trials	Normalized MSE	Averaged DTW
Random	Visual	Midbrain	2960	0.9463	0.3493
	Visual	Hippocampus	4424	1.1897	0.3526
	Visual	Basal Ganglia	2279	1.2185	0.8020
Difference between max and min	Visual	Midbrain	2960	1.1138	0.3928
	Visual	Hippocampus	4424	1.0714	0.2770
	Visual	Basal Ganglia	2279	1.6242	0.6530
Standard deviation	Visual	Midbrain	2960	1.0647	1.3156
	Visual	Hippocampus	4424	2.2245	0.3217
	Visual	Basal Ganglia	2279	1.0333	0.7464

Table 3 shows that performance across selection methods varies inconsistently by region. While we hypothesized that midbrain activity would be easier to predict, these results provide only limited support. The patterns of normalized MSE and DTW across different neuron selection methods are inconsistent and are considerably sensitive to the quality of input neurons, suggesting that more targeted identification of choice-selective neurons in both the source and target regions may be necessary for meaningful comparison. Future work could explore data-driven selection strategies,

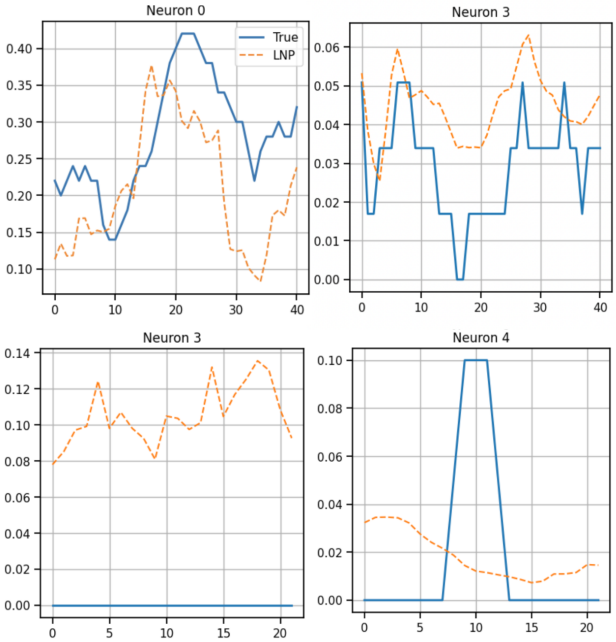


Figure 4: Performance across some testing neurons. The top row shows examples of accurate prediction, whereas the bottom row reflects mispredictions.

such as PCA or d-prime-based clustering, to better localize informative neurons. Also, constructing a more informative evaluation metrics may be beneficial to combine both errors and shapes.

Despite the high variability in the training dataset, the model was still able to capture some meaningful patterns in neural activity. This indicates that leveraging temporal-spectral representations for neural time-series prediction is a possible direction for modeling complex, distributed brain dynamics.

5 Conclusion

Overall, our analyses demonstrated that both ROI-level and neuron-level models can predict downstream brain activity from upstream regions. Different predictive performance of the models could provide insights into functional connectivity. Population-level GLMs revealed structured inputs to the midbrain and hippocampus, while the basal ganglia showed more variability. Neuron-level models captured meaningful temporal patterns in some cases, despite session-level variability. Together, these findings highlight the promise and limitations of modeling brain-wide information flow at multiple scales.

6 Code

The data and code of our analysis can be found at this link:

https://drive.google.com/drive/folders/19Cm57w7k90LgeNVlwRhwr3UMb56vHUm2?usp=sharing.

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7 Appendix

Table 4: Summary of task condition coefficients across all models. Statistically significant coefficients ($p < 0.05$) are marked with *.

Model Type	Target	Contrast Left	Contrast Right	No-Go
Poisson (Task Only)	Midbrain	-0.0021 (0.740)	-0.0247* (0.000)	-0.2199* (0.000)
Poisson (Task Only)	Hippocampus	-0.0209* (0.012)	0.0542* (0.000)	-0.0936* (0.000)
Poisson (Task Only)	Basal Ganglia	0.0237* (0.015)	0.0092 (0.367)	-0.2187* (0.000)
Poisson (Full Brain Lasso)	Midbrain	-0.0199* (0.003)	-0.0390* (0.000)	-0.1608* (0.000)
Poisson (Full Brain Lasso)	Hippocampus	-0.0604* (0.000)	0.0296* (0.001)	-0.1132* (0.000)
Poisson (Full Brain Lasso)	Basal Ganglia	0.0092 (0.353)	0.0337* (0.001)	-0.2615* (0.000)
Poisson (Visual Lasso)	Midbrain	-0.0038 (0.535)	-0.0186* (0.003)	-0.2198* (0.000)
Poisson (Visual Lasso)	Hippocampus	-0.0508* (0.000)	0.0472* (0.000)	-0.1138* (0.000)
Poisson (Visual Lasso)	Basal Ganglia	0.0380* (0.000)	0.0480* (0.000)	-0.2782* (0.000)
NB (Full Brain Lasso)	Midbrain	-0.0070 (0.866)	-0.0479 (0.257)	-0.0896* (0.020)
NB (Full Brain Lasso)	Hippocampus	-0.0511 (0.143)	0.0288 (0.426)	-0.1169* (0.000)
NB (Full Brain Lasso)	Basal Ganglia	-0.0373 (0.380)	0.0066 (0.883)	-0.2708* (0.000)
NB (Visual Lasso)	Midbrain	-0.0040 (0.923)	-0.0079 (0.850)	-0.1795* (0.000)
NB (Visual Lasso)	Hippocampus	0.5977 (0.102)	0.0446 (0.213)	-0.1077* (0.000)
NB (Visual Lasso)	Basal Ganglia	0.003 (0.995)	0.0210 (0.637)	-0.2898* (0.000)

Table 5: Summary of significant predictors from visual and thalamic inputs for target areas (NB GLM).

Target Region	**R^2**	**Significant ROIs ($p < 0.05$)**	**Top Positive Coefficients**
Midbrain	0.115	VISa, VISam, VISrl, LH, MD, POL, VPL	**MD** (+7.42) – A higher-order thalamic nucleus involved in relaying motor-related sensory signals; suggests strong top-down influence on midbrain dynamics. **LH** (+2.65) – Lateral habenula is involved in reward and aversion processing; may contribute to midbrain modulation during goal-directed behavior. **VPL** (+1.47) – Primary somatosensory thalamic relay; indicates midbrain integrates tactile/motor state feedback early in the pathway.
Hippocampus	0.148	VISp, VISrl, CL, LGd, LH, POL, PT, VPL	**PT** (+0.41) – Posterior thalamic nucleus linked to multisensory integration; suggests contextual sensory information shapes hippocampal dynamics. **VPL** (+0.36) – Reinforces the idea that somatosensory inputs modulate hippocampal encoding of sensorimotor contingencies. **LGd** (+0.28) – Visual thalamic relay; suggests visual context contributes to spatial/mnemonic processing in hippocampus.
Basal Ganglia	0.039	VISp, VISp, VISrl, LD, LP, MD, VAL	**MD** (+0.22) – Mediodorsal thalamus participates in prefrontal-basal ganglia loops; indicates cognitive context influences basal ganglia output. **LD** (+0.15) – Lateral dorsal thalamus has hippocampal connections; may channel spatial or memory-related context into BG circuits. **LP** (+0.13) – Visual-association thalamic region; supports idea that action selection is informed by high-level visual cues.

Table 6: Summary of significant predictors from midbrain inputs for target areas (NB GLM).

Target Region	**R^2**	**Significant ROIs ($p < 0.05$)**	**Top Positive Coefficients**
Hippocampus	0.149	APN, IC, MRN, NB, RN, SCs, SCm, SCsg, ZI	**SCm** (+0.98) – Medial superior colliculus; may relay visual-spatial attention signals to hippocampus during navigation or target selection. **RN** (+0.64) – Red nucleus; conveys motor coordination signals, potentially influencing hippocampal representations linked to planned movements. **SCsg** (+0.38) – Superficial gray layer of SC; integrates early visual salience information relevant for episodic encoding.
Basal Ganglia	0.017	APN, MRN, PAG, RN, SCs, SCm, SCig	**SCs** (+0.73) – Sensory layer of SC; contributes to orienting responses and may influence BG action selection circuits. **PAG** (+0.41) – Periaqueductal gray; involved in defensive behaviors and motivational states, possibly gating motor output through BG. **MRN** (+0.51) – Midbrain reticular nucleus; modulates arousal and state-dependent processing that could affect BG readiness or suppression.